

Simple random and systematic sampling

SAMPLING IN PYTHON



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Simple random sampling



Simple random sampling of coffees



Simple random sampling with pandas

```
coffee_ratings.sample(n=5, random_state=19000113)
```

	total_cup_points		variety	country_of_origin	aroma	flavor	\
437	83.25		None	Colombia	7.92	7.75	
285	83.83	Yellow	Bourbon	Brazil	7.92	7.50	
784	82.08		None	Colombia	7.50	7.42	
648	82.58		Caturra	Colombia	7.58	7.50	
155	84.58		Caturra	Colombia	7.42	7.67	

	aftertaste	body	balance
437	7.25	7.83	7.58
285	7.33	8.17	7.50
784	7.42	7.67	7.42
648	7.42	7.67	7.42
155	7.75	8.08	7.83

Systematic sampling



Systematic sampling - defining the interval

```
sample_size = 5  
pop_size = len(coffee_ratings)  
print(pop_size)
```

1338

```
interval = pop_size // sample_size  
print(interval)
```

267

Systematic sampling - selecting the rows

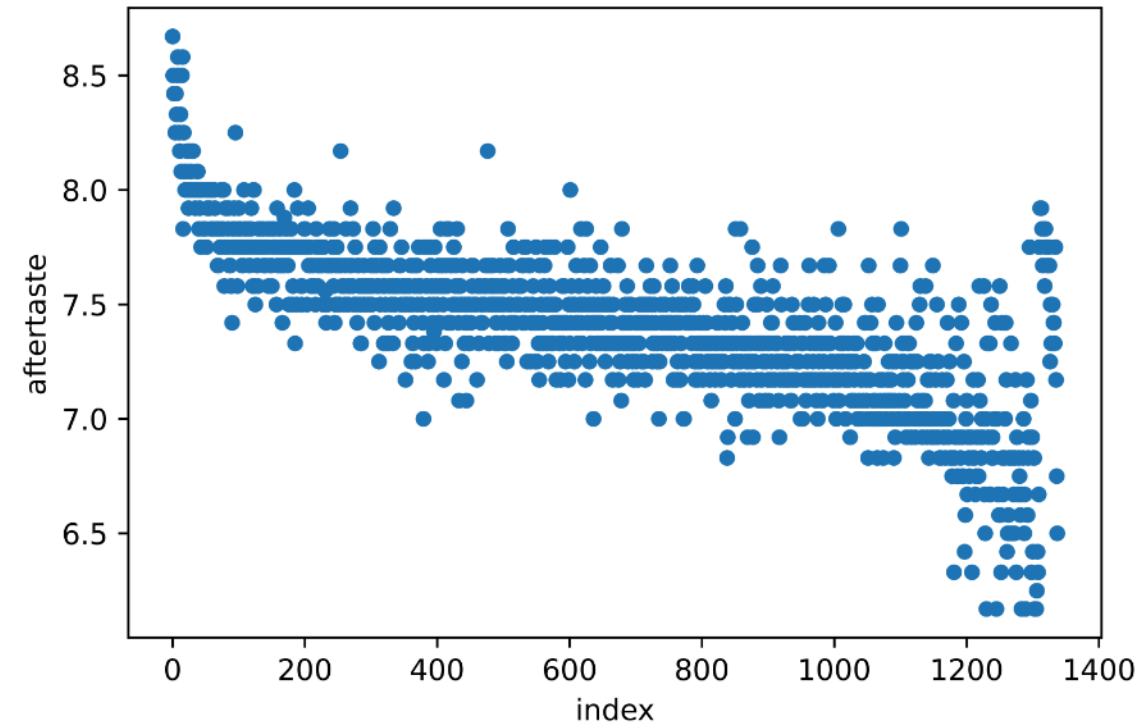
```
coffee_ratings.iloc[::interval]
```

	total_cup_points	variety	country_of_origin	aroma	flavor	aftertaste	\
0	90.58	None	Ethiopia	8.67	8.83	8.67	
267	83.92	None	Colombia	7.83	7.75	7.58	
534	82.92	Bourbon	El Salvador	7.50	7.50	7.75	
801	82.00	Typica	Taiwan	7.33	7.50	7.17	
1068	80.50	Other	Taiwan	7.17	7.17	7.17	

	body	balance
0	8.50	8.42
267	7.75	7.75
534	7.92	7.83
801	7.50	7.33
1068	7.17	7.25

The trouble with systematic sampling

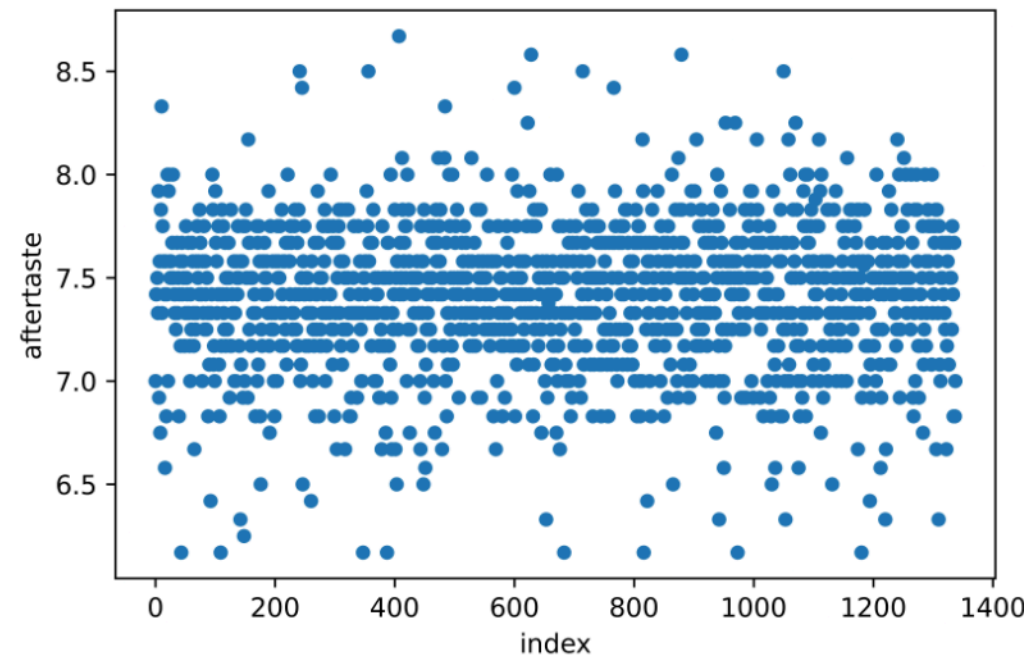
```
coffee_ratings_with_id = coffee_ratings.reset_index()
coffee_ratings_with_id.plot(x="index", y="aftertaste", kind="scatter")
plt.show()
```



Systematic sampling is only safe if we don't see a pattern in this scatter plot

Making systematic sampling safe

```
shuffled = coffee_ratings.sample(frac=1)
shuffled = shuffled.reset_index(drop=True).reset_index()
shuffled.plot(x="index", y="aftertaste", kind="scatter")
plt.show()
```



Shuffling rows + systematic sampling is the same as simple random sampling

Let's practice!

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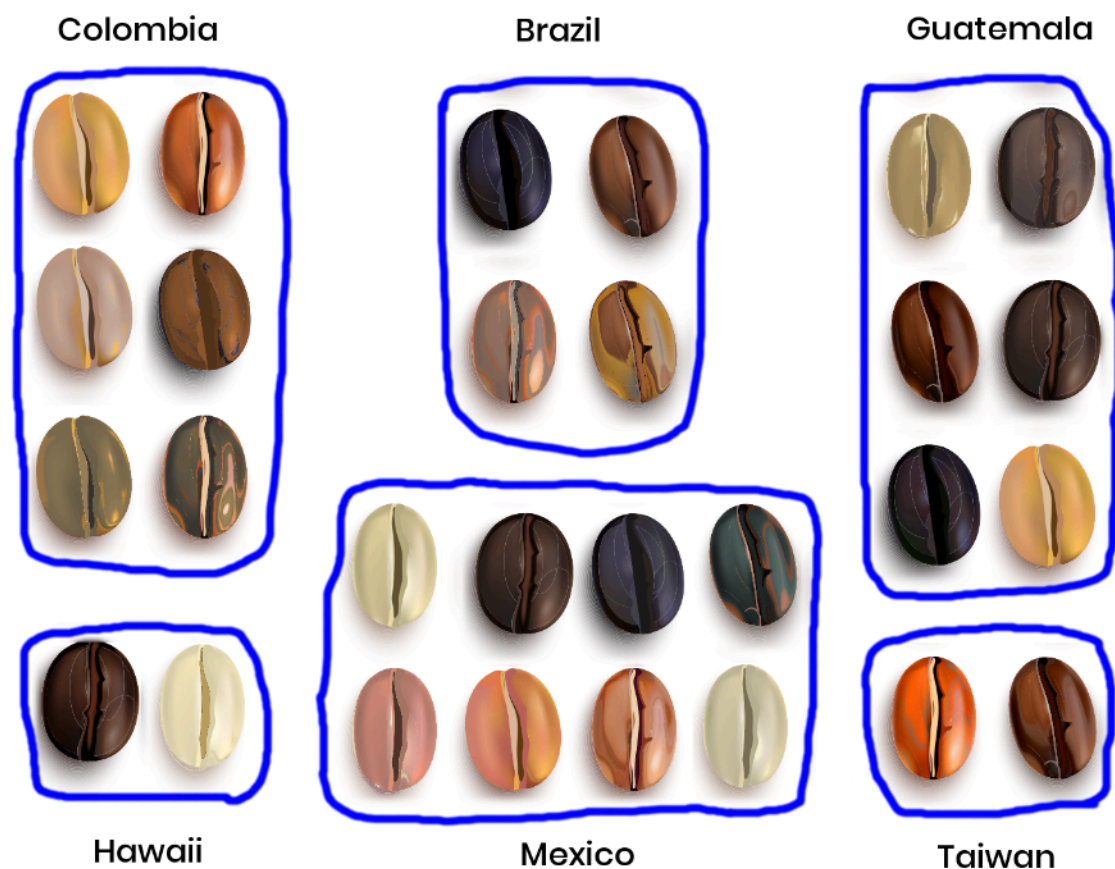
Stratified and weighted random sampling

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Coffees by country



```
top_counts = coffee_ratings['country_of_origin'].value_counts()
top_counts.head(6)
```

```
country_of_origin
Mexico                236
Colombia              183
Guatemala            181
Brazil               132
Taiwan                75
United States (Hawaii) 73
dtype: int64
```

¹ The dataset lists Hawaii and Taiwan as countries for convenience, as they are notable coffee-growing regions.

Filtering for 6 countries

```
top_counted_countries = ["Mexico", "Colombia", "Guatemala",  
    "Brazil", "Taiwan", "United States (Hawaii)"]  
  
top_counted_subset = coffee_ratings['country_of_origin'].isin(top_counted_countries)  
  
coffee_ratings_top = coffee_ratings[top_counted_subset]
```

Counts of a simple random sample

```
coffee_ratings_samp = coffee_ratings_top.sample(frac=0.1, random_state=2021)
```

```
coffee_ratings_samp['country_of_origin'].value_counts(normalize=True)
```

```
country_of_origin
Mexico           0.250000
Guatemala        0.204545
Colombia          0.181818
Brazil           0.181818
United States (Hawaii) 0.102273
Taiwan           0.079545
dtype: float64
```

Comparing proportions

Population:

```
Mexico          0.268182
Colombia         0.207955
Guatemala        0.205682
Brazil           0.150000
Taiwan           0.085227
United States (Hawaii) 0.082955
Name: country_of_origin, dtype: float64
```

10% simple random sample:

```
Mexico          0.250000
Guatemala        0.204545
Colombia         0.181818
Brazil           0.181818
United States (Hawaii) 0.102273
Taiwan           0.079545
Name: country_of_origin, dtype: float64
```

Proportional stratified sampling

```
coffee_ratings_strat = coffee_ratings_top.groupby("country_of_origin")\
    .sample(frac=0.1, random_state=2021)
```

```
coffee_ratings_strat['country_of_origin'].value_counts(normalize=True)
```

```
Mexico          0.272727
Guatemala       0.204545
Colombia        0.204545
Brazil          0.147727
Taiwan          0.090909
United States (Hawaii) 0.079545
Name: country_of_origin, dtype: float64
```

Equal counts stratified sampling

```
coffee_ratings_eq = coffee_ratings_top.groupby("country_of_origin")\
    .sample(n=15, random_state=2021)
```

```
coffee_ratings_eq['country_of_origin'].value_counts(normalize=True)
```

```
Taiwan          0.166667
Brazil          0.166667
United States (Hawaii)  0.166667
Guatemala       0.166667
Mexico          0.166667
Colombia        0.166667
Name: country_of_origin, dtype: float64
```

Weighted random sampling

- Specify weights to adjust the relative probability of a row being sampled

```
import numpy as np
coffee_ratings_weight = coffee_ratings_top
condition = coffee_ratings_weight['country_of_origin'] == "Taiwan"
```

```
coffee_ratings_weight['weight'] = np.where(condition, 2, 1)
```

```
coffee_ratings_weight = coffee_ratings_weight.sample(frac=0.1, weights="weight")
```

Weighted random sampling results

10% weighted sample:

```
coffee_ratings_weight['country_of_origin'].value_counts(normalize=True)
```

```
Brazil          0.261364
Mexico          0.204545
Guatemala       0.204545
Taiwan          0.170455
Colombia        0.090909
United States (Hawaii) 0.068182
Name: country_of_origin, dtype: float64
```

Let's practice!

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Cluster sampling

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Stratified sampling vs. cluster sampling

Stratified sampling

- Split the population into subgroups
- Use simple random sampling on every subgroup

Cluster sampling

- Use simple random sampling to pick some subgroups
- Use simple random sampling on only those subgroups

Varieties of coffee



```
varieties_pop = list(coffee_ratings['variety'].unique())
```

```
[None, 'Other', 'Bourbon', 'Catimor',  
'Ethiopian Yirgacheffe', 'Caturra',  
'SL14', 'Sumatra', 'SL34', 'Hawaiian Kona',  
'Yellow Bourbon', 'SL28', 'Gesha', 'Catuai',  
'Pacamara', 'Typica', 'Sumatra Lintong',  
'Mundo Novo', 'Java', 'Peaberry', 'Pacas',  
'Mandheling', 'Ruiru 11', 'Arusha',  
'Ethiopian Heirloom', 'Moka Peaberry',  
'Sulawesi', 'Blue Mountain', 'Marigojipe',  
'Pache Comun']
```

Stage 1: sampling for subgroups



```
import random  
varieties_samp = random.sample(varieties_pop, k=3)
```

```
['Hawaiian Kona', 'Bourbon', 'SL28']
```

Stage 2: sampling each group

```
variety_condition = coffee_ratings['variety'].isin(varieties_samp)
coffee_ratings_cluster = coffee_ratings[variety_condition]
```

```
coffee_ratings_cluster['variety'] = coffee_ratings_cluster['variety'].cat.remove_unused_categories()
```

```
coffee_ratings_cluster.groupby("variety")\
    .sample(n=5, random_state=2021)
```

Stage 2 output

```
total_cup_points    variety    country_of_origin ...
variety
Bourbon      575      82.83    Bourbon      Guatemala
              560      82.83    Bourbon      Guatemala
              524      83.00    Bourbon      Guatemala
              1140     79.83    Bourbon      Guatemala
              318      83.67    Bourbon      Brazil
Hawaiian Kona 1291     73.67    Hawaiian Kona United States (Hawaii)
              1266     76.25    Hawaiian Kona United States (Hawaii)
              488      83.08    Hawaiian Kona United States (Hawaii)
              461      83.17    Hawaiian Kona United States (Hawaii)
              117      84.83    Hawaiian Kona United States (Hawaii)
SL28         137      84.67    SL28         Kenya
              452      83.17    SL28         Kenya
              224      84.17    SL28         Kenya
              66       85.50    SL28         Kenya
              559      82.83    SL28         Kenya
```

Multistage sampling

- Cluster sampling is a type of multistage sampling
- Can have > 2 stages
- E.g., countrywide surveys may sample states, counties, cities, and neighborhoods

Let's practice!

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Comparing sampling methods

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Review of sampling techniques - setup

```
top_counted_countries = ["Mexico", "Colombia", "Guatemala",  
    "Brazil", "Taiwan", "United States (Hawaii)"]  
  
subset_condition = coffee_ratings['country_of_origin'].isin(top_counted_countries)  
coffee_ratings_top = coffee_ratings[subset_condition]  
  
coffee_ratings_top.shape
```

```
(880, 8)
```

Review of simple random sampling

```
coffee_ratings_srs = coffee_ratings_top.sample(frac=1/3, random_state=2021)
```

```
coffee_ratings_srs.shape
```

```
(293, 8)
```

Review of stratified sampling

```
coffee_ratings_strat = coffee_ratings_top.groupby("country_of_origin")\
    .sample(frac=1/3, random_state=2021)
```

```
coffee_ratings_strat.shape
```

```
(293, 8)
```

Review of cluster sampling

```
import random
top_countries_samp = random.sample(top_counted_countries, k=2)
top_condition = coffee_ratings_top['country_of_origin'].isin(top_countries_samp)
coffee_ratings_cluster = coffee_ratings_top[top_condition]
coffee_ratings_cluster['country_of_origin'] = coffee_ratings_cluster['country_of_origin']\
    .cat.remove_unused_categories()

coffee_ratings_clust = coffee_ratings_cluster.groupby("country_of_origin")\
    .sample(n=len(coffee_ratings_top) // 6)
```

```
coffee_ratings_clust.shape
```

```
(292, 8)
```

Calculating mean cup points

Population

```
coffee_ratings_top['total_cup_points'].mean()
```

```
81.947000000000002
```

Simple random sample

```
coffee_ratings_srs['total_cup_points'].mean()
```

```
81.95982935153583
```

Stratified sample

```
coffee_ratings_strat['total_cup_points'].mean()
```

```
81.92566552901025
```

Cluster sample

```
coffee_ratings_clust['total_cup_points'].mean()
```

```
82.03246575342466
```

Mean cup points by country: simple random

Population:

```
coffee_ratings_top.groupby("country_of_origin")\
    ['total_cup_points'].mean()
```

```
country_of_origin
Brazil           82.405909
Colombia         83.106557
Guatemala       81.846575
Mexico          80.890085
Taiwan          82.001333
United States (Hawaii) 81.820411
Name: total_cup_points, dtype: float64
```

Simple random sample:

```
coffee_ratings_srs.groupby("country_of_origin")\
    ['total_cup_points'].mean()
```

```
country_of_origin
Brazil           82.414878
Colombia         82.925536
Guatemala       82.045385
Mexico          81.100714
Taiwan          81.744333
United States (Hawaii) 82.008000
Name: total_cup_points, dtype: float64
```

Mean cup points by country: stratified

Population:

```
coffee_ratings_top.groupby("country_of_origin")\
    ['total_cup_points'].mean()
```

```
country_of_origin
Brazil           82.405909
Colombia         83.106557
Guatemala        81.846575
Mexico           80.890085
Taiwan           82.001333
United States (Hawaii) 81.820411
Name: total_cup_points, dtype: float64
```

Stratified sample:

```
coffee_ratings_strat.groupby("country_of_origin")\
    ['total_cup_points'].mean()
```

```
country_of_origin
Brazil           82.499773
Colombia         83.288197
Guatemala        81.727667
Mexico           80.994684
Taiwan           81.846800
United States (Hawaii) 81.051667
Name: total_cup_points, dtype: float64
```

Mean cup points by country: cluster

Population:

```
coffee_ratings_top.groupby("country_of_origin")\
    ['total_cup_points'].mean()
```

```
country_of_origin
Brazil           82.405909
Colombia         83.106557
Guatemala       81.846575
Mexico          80.890085
Taiwan          82.001333
United States (Hawaii) 81.820411
Name: total_cup_points, dtype: float64
```

Cluster sample:

```
coffee_ratings_clust.groupby("country_of_origin")\
    ['total_cup_points'].mean()
```

```
country_of_origin
Colombia      83.128904
Mexico       80.936027
Name: total_cup_points, dtype: float64
```

Let's practice!

SAMPLING IN PYTHON

